


Each cross-layer explicitly creates high-order feature interactions

$$x_0 = [a, b, c]$$

a - user age

b - item price

c - device type (smartphone, TV, laptop, web)

Linear: $\text{score} = w_1 a + w_2 b + w_3 c$

↑
↑
↑
separately!

DCN

$$[x_0^0 \cdot c, x_0^1 \cdot c, \dots, x_0^n \cdot c]$$

V1: $x_{e+1} = x_0 (w_e^T x_e) + b_e + x_e$
constant = c

V2: $x_{e+1} = x_0 \odot (w_e^T x_e + b_e) + x_e$
[$x_0^0 \cdot c^0, x_0^1 \cdot c^1, \dots, x_0^n \cdot c^n$]

Ⓟ $x_1 = x_0 (w_0^T x_0) + x_0, \quad w_0 = [w_a, w_b, w_c]$

$$x_0 \cdot w_0^T x_0 = w_a a + w_b b + w_c c$$

$$x_0 (w_0^T x_0) = [a, b, c] (w_a a + w_b b + w_c c)$$

Expanding: $(w_a a^2 + w_a a b + w_a a c, w_b a b + w_b b^2 + w_b b c, w_c a c + w_c b c + w_c c^2)$

After next cross layer: $a^3, b^3, c^3, a^2 b, a^2 c, b^2 c, b^2 a, c^2 a, c^2 b, a, b, c$